

Python Quiz

Question 1: Basic Definitions

Q1: Define the following terms in Python:

- a) Function
- b) Class
- c) Module
- d) List
- e) Dictionary

Question 2: Functions and Classes

Q2: Write a Python function that takes a list of numbers and returns a dictionary with the count of even and odd numbers. Then, create a class that uses this function.

Question 3: NumPy Operations

Q3: Using NumPy, write a program to create a 3x3 matrix and perform the following operations:

- a) Find the determinant.
- b) Find the inverse.
- c) Calculate the mean of the elements.

Question 4: Set Operations

Q4: Write a Python program to demonstrate the following set operations:

- a) Union
- b) Intersection
- c) Difference
- d) Symmetric difference

Question 5: Lists and List Comprehension

Q5: Write a Python program to:

- a) Create a list of the first 10 even numbers.
- b) Use list comprehension to create a new list with the squares of the even numbers.
- c) Filter the new list to include only numbers greater than 50.

Example:

```
even_numbers = [2 * i for i in range(1, 11)]
```

```
squares = [x**2 for x in even_numbers]
```

```
filtered_squares = [x for x in squares if x > 50]
```